

(12) **United States Patent**
Enquist et al.

(10) **Patent No.:** **US 12,401,011 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **STACKED DEVICES AND METHODS OF FABRICATION**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 398 days.

(21) Appl. No.: **18/148,351**

(22) Filed: **Dec. 29, 2022**

(65) **Prior Publication Data**

US 2023/0282634 A1 Sep. 7, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/586,236, filed on Jan. 27, 2022, now Pat. No. 11,916,054, which is a (Continued)

(51) **Int. Cl.**
H01L 25/18 (2023.01)
H01L 21/768 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H01L 25/18** (2013.01); **H01L 21/76898** (2013.01); **H01L 23/3171** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC H01L 23/481; H01L 2225/06541-06544; H01L 2225/06548; H01L 21/76898;
(Continued)

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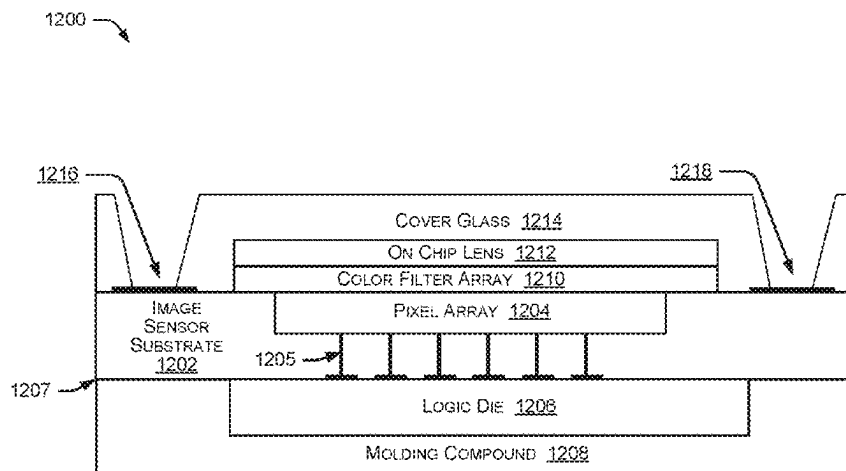
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(57) **ABSTRACT**

Stacked devices and methods of fabrication are provided. Die-to-wafer (D2W) direct-bonding techniques join layers of dies of various physical sizes, form factors, and foundry nodes to a semiconductor wafer, to interposers, or to boards and panels, allowing mixing and matching of variegated dies in the fabrication of 3D stacked devices during wafer level packaging (WLP). Molding material fills in lateral spaces between dies to enable fan-out versions of 3D die stacks with fine pitch leads and capability of vertical through-vias throughout. Molding material is planarized to create direct-bonding surfaces between multiple layers of the variegated dies for high interconnect density and reduction of vertical height. Interposers with variegated dies on one or both sides can be created and bonded to wafers. Logic dies and image sensors from different fabrication nodes and different wafer sizes can be stacked during WLP, or logic dies and high bandwidth memory (HBM) of different geometries can be stacked during WLP.

28 Claims, 13 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/413,429, filed on May 15, 2019, now Pat. No. 11,276,676.

- (60) Provisional application No. 62/671,917, filed on May 15, 2018.

(51) **Int. Cl.**

H01L 23/00 (2006.01)

H01L 23/31 (2006.01)

H01L 23/48 (2006.01)

H01L 25/00 (2006.01)

H01L 25/07 (2006.01)

H10D 84/03 (2025.01)

H10D 88/00 (2025.01)

H10F 39/00 (2025.01)

(52) **U.S. Cl.**

CPC **H01L 23/481** (2013.01); **H01L 24/09** (2013.01); **H01L 25/074** (2013.01); **H01L 25/50** (2013.01); **H10D 84/038** (2025.01); **H10D 88/01** (2025.01); **H10F 39/018** (2025.01); **H10F 39/026** (2025.01); **H10F 39/804** (2025.01); **H10F 39/8053** (2025.01); **H10F 39/8063** (2025.01); **H10F 39/809** (2025.01); **H10F 39/811** (2025.01); **H01L 2224/02379** (2013.01); **H01L 2924/1431** (2013.01); **H01L 2924/1434** (2013.01)

(58) **Field of Classification Search**

CPC **H01L 23/28–3192**; **H01L 21/56–568**; **H01L 2924/181–186**; **H01L 21/187**; **H01L 2224/80894–80896**; **H01L 2224/81894–81896**; **H01L 2224/82895–82897**; **H01L 2224/83894–83896**; **H10F 39/804**

See application file for complete search history.

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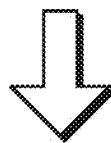
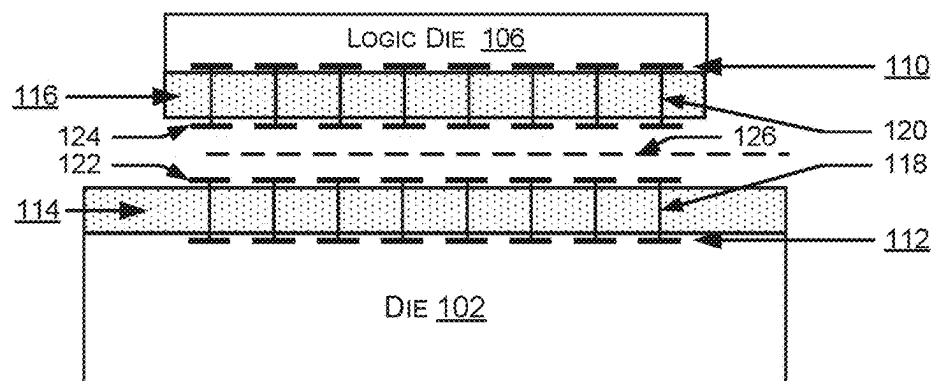
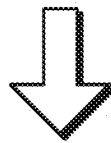
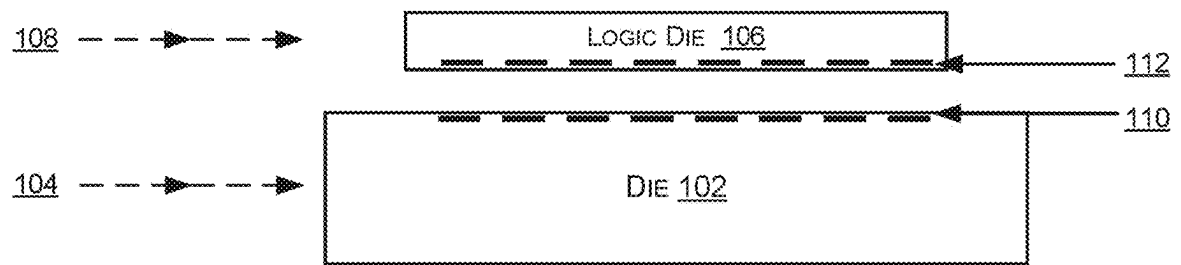


FIG. 1

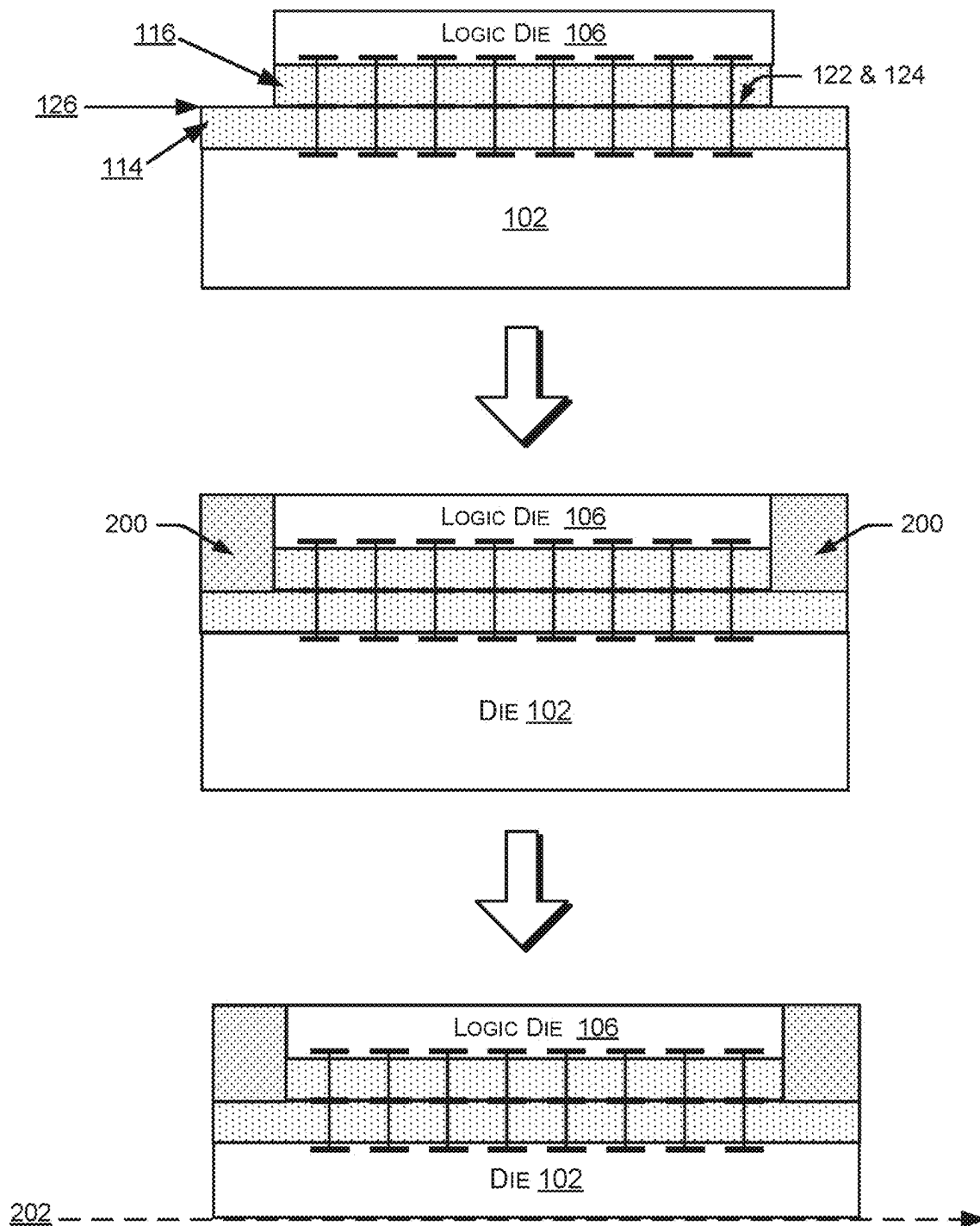


FIG. 2

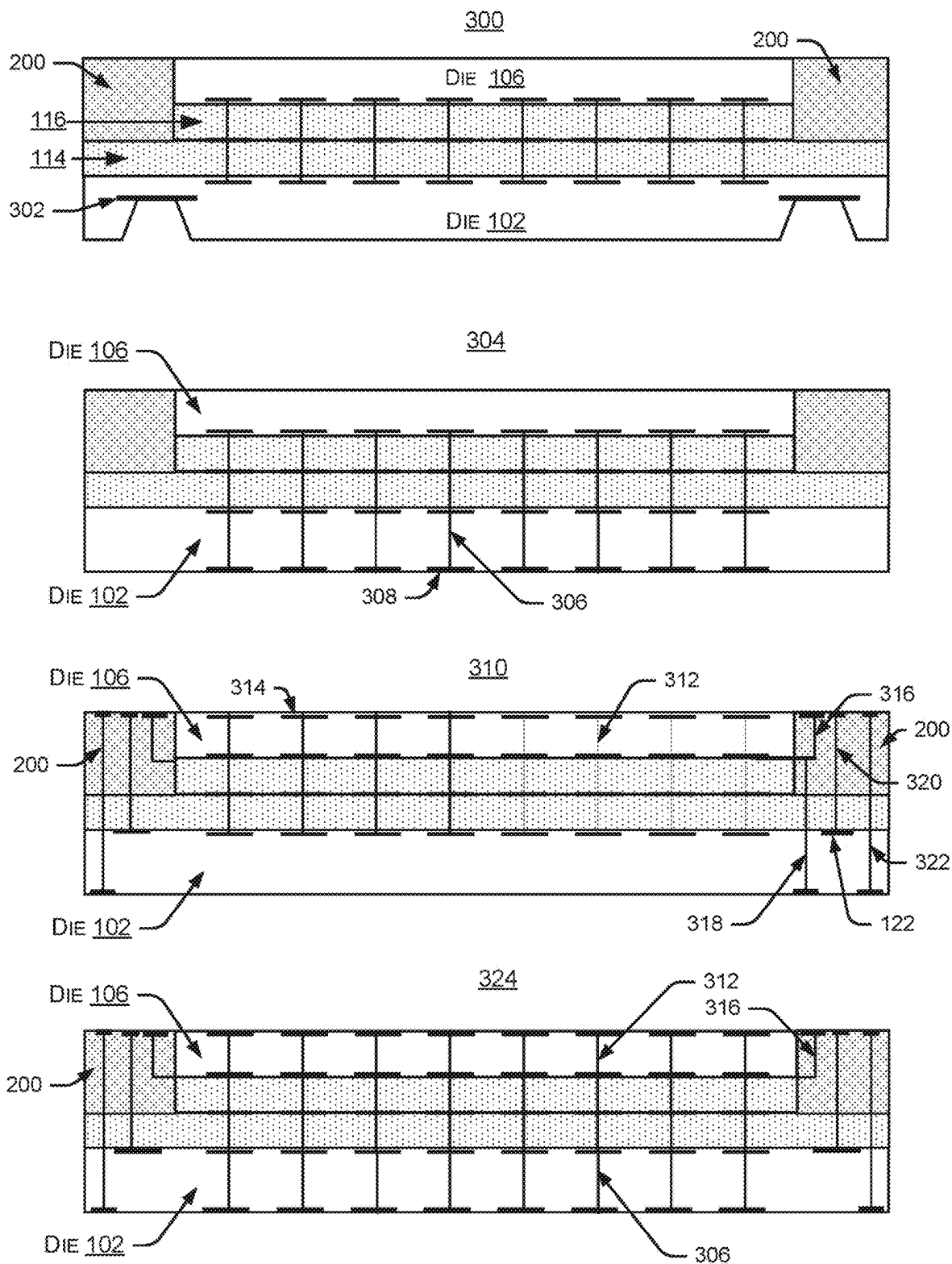


FIG. 3

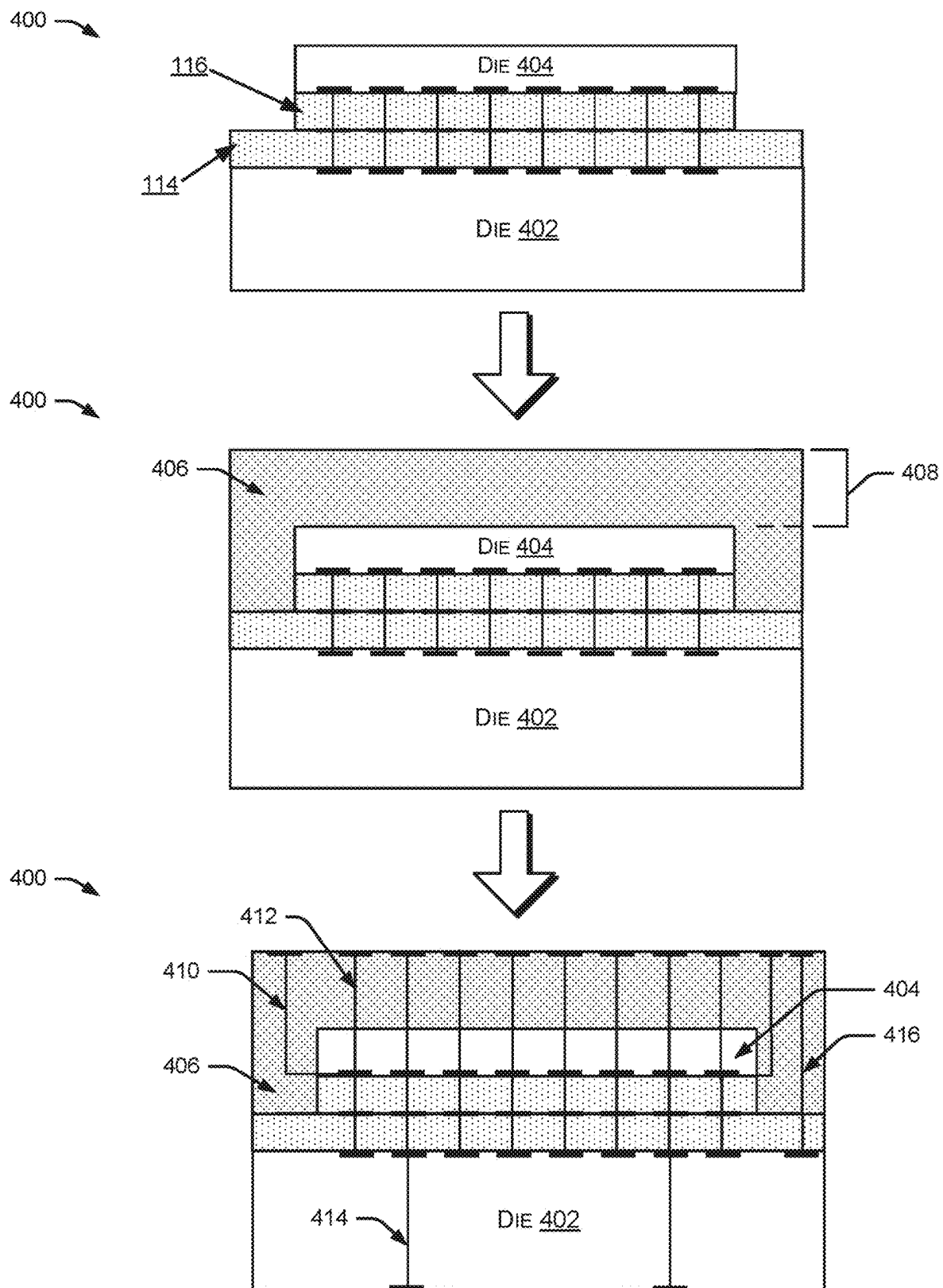


FIG. 4

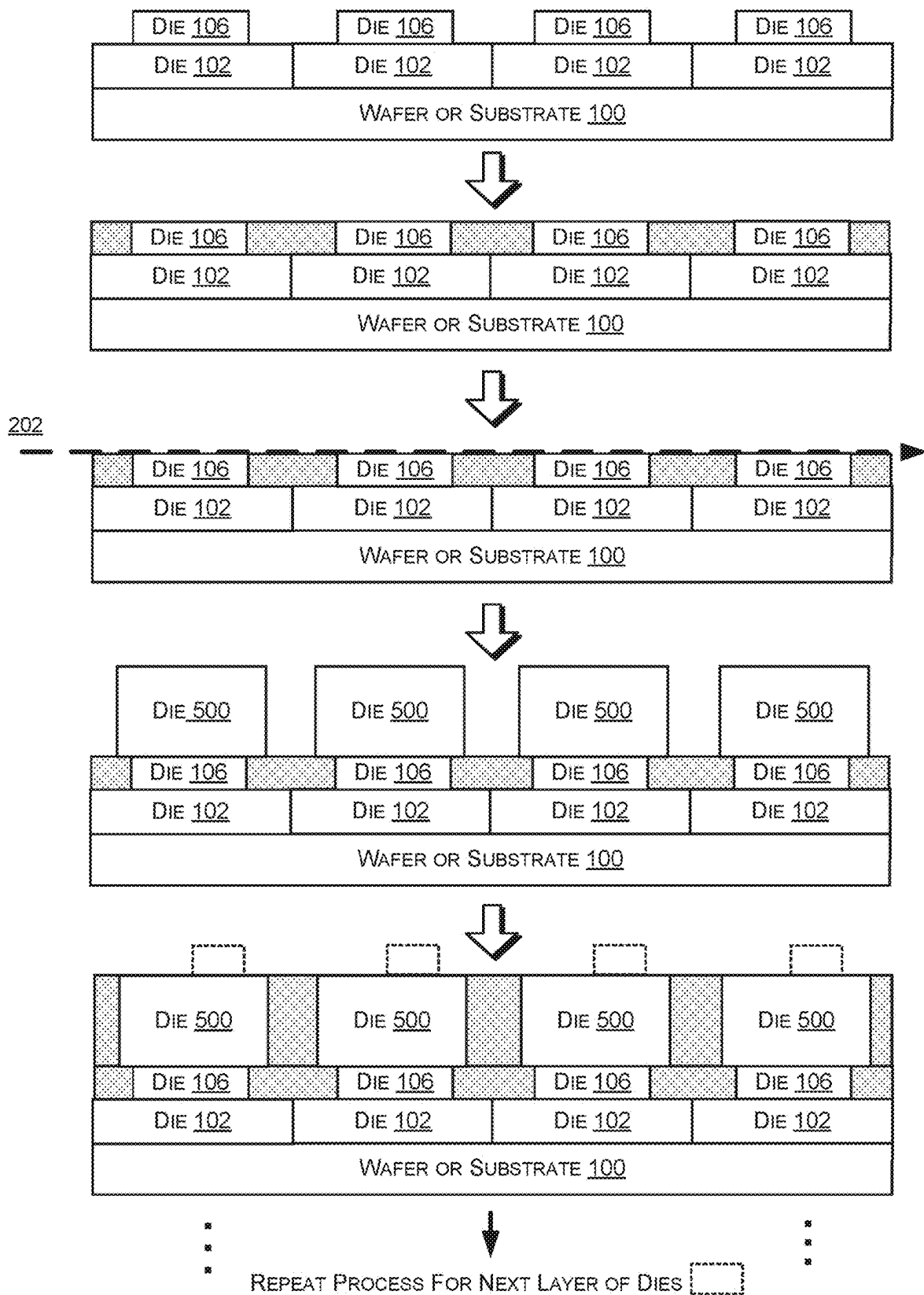
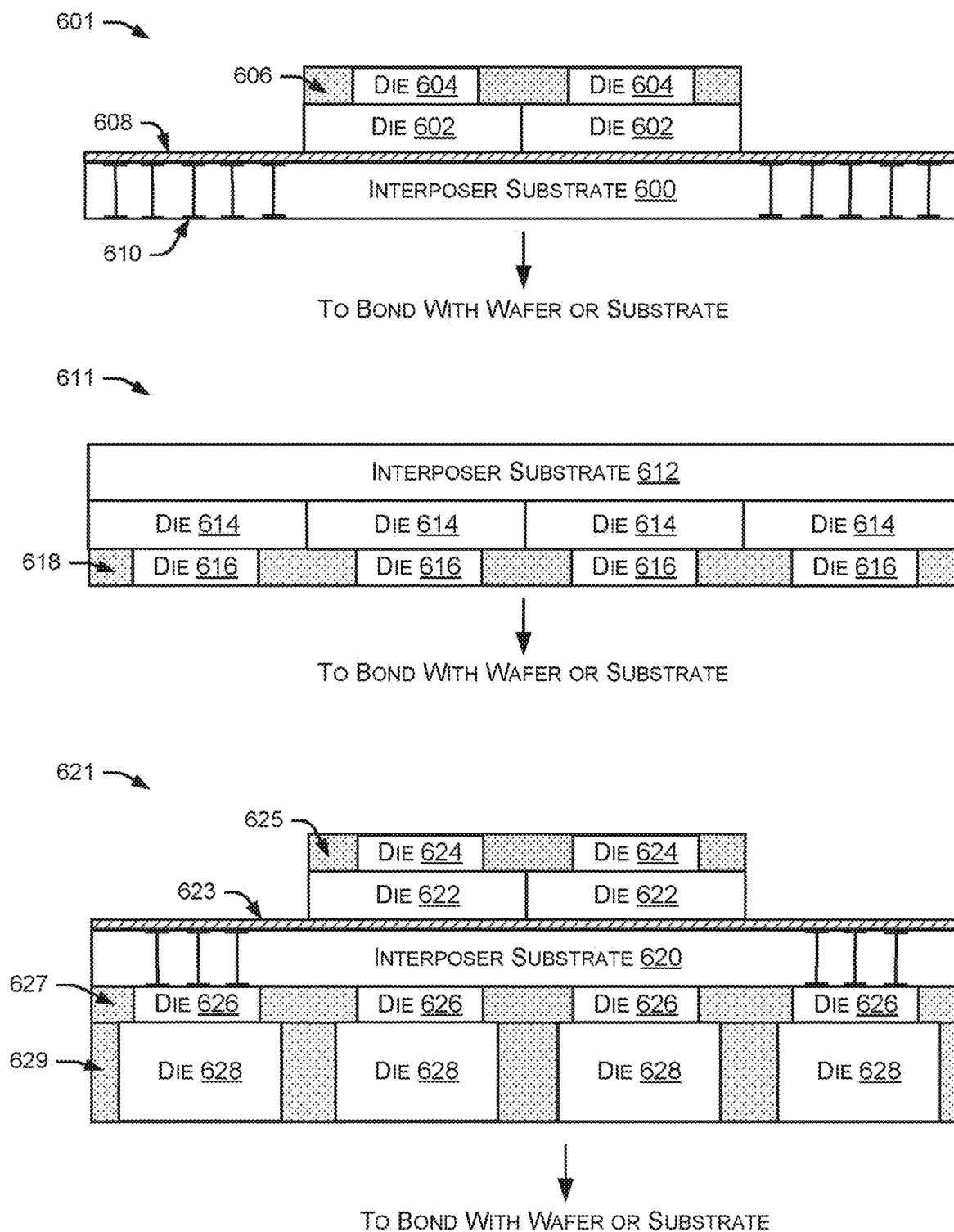


FIG. 5

**FIG. 6**

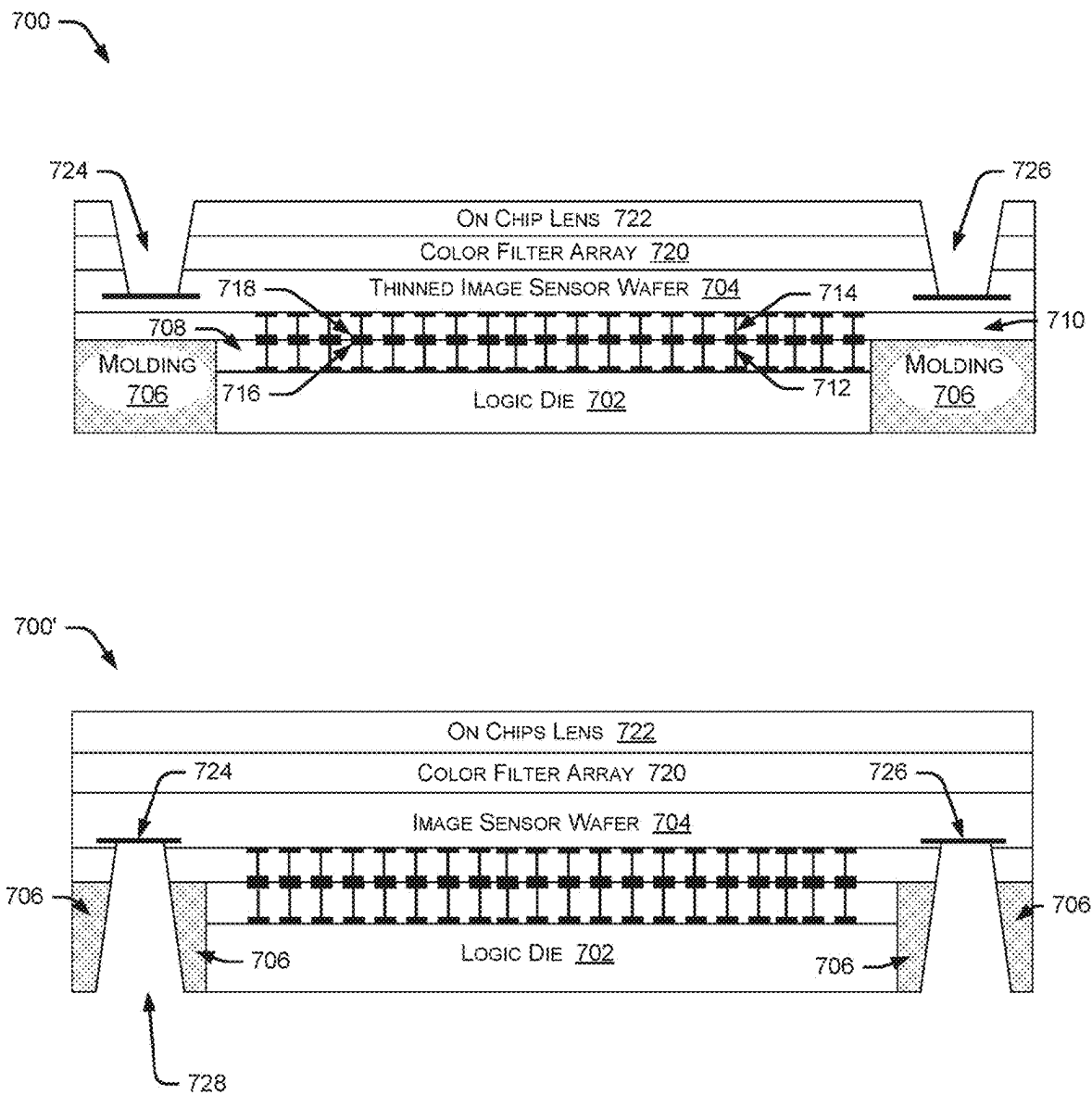
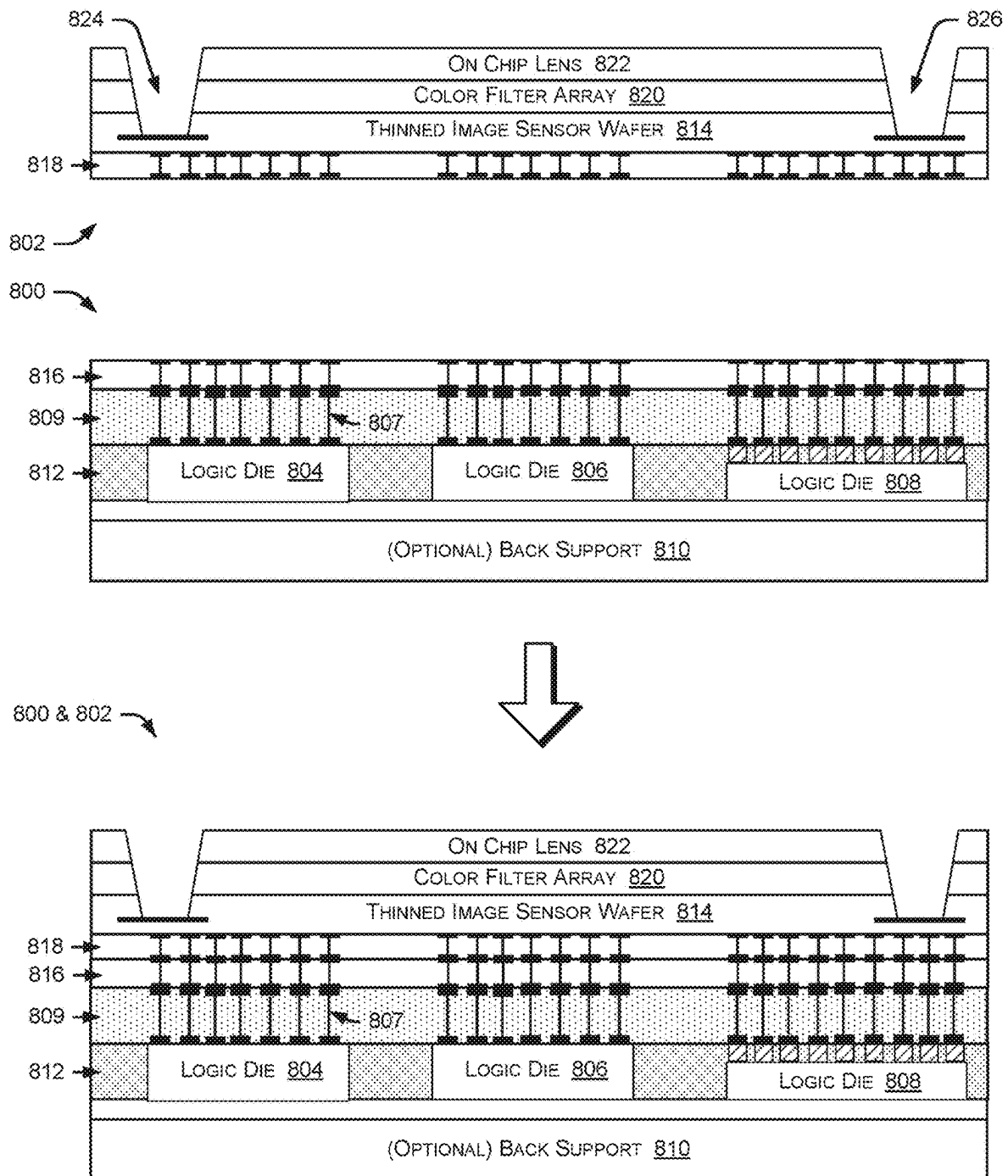
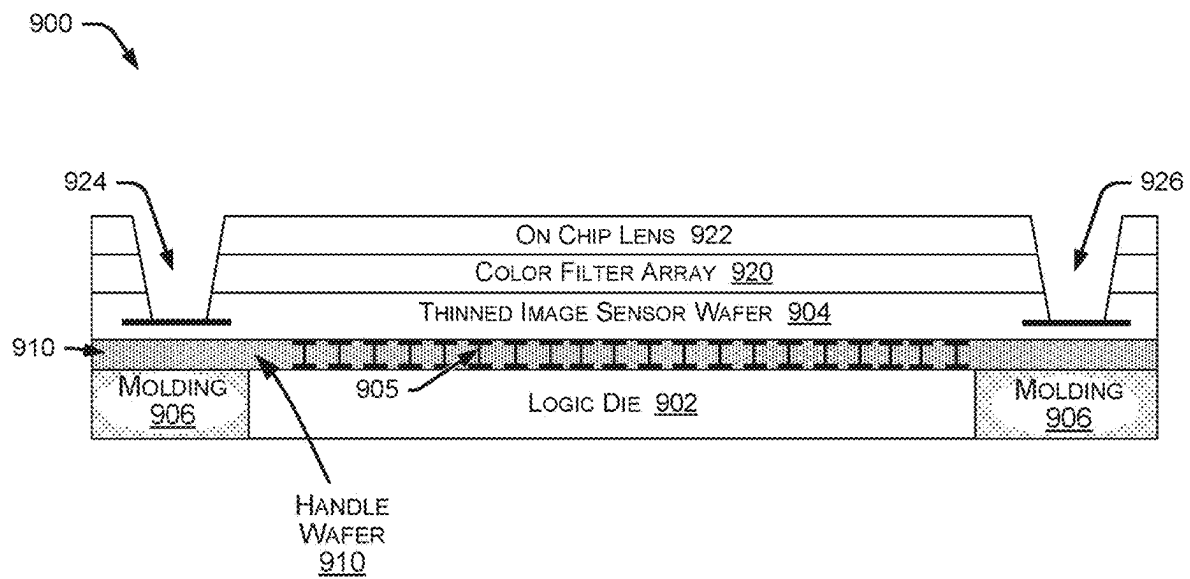
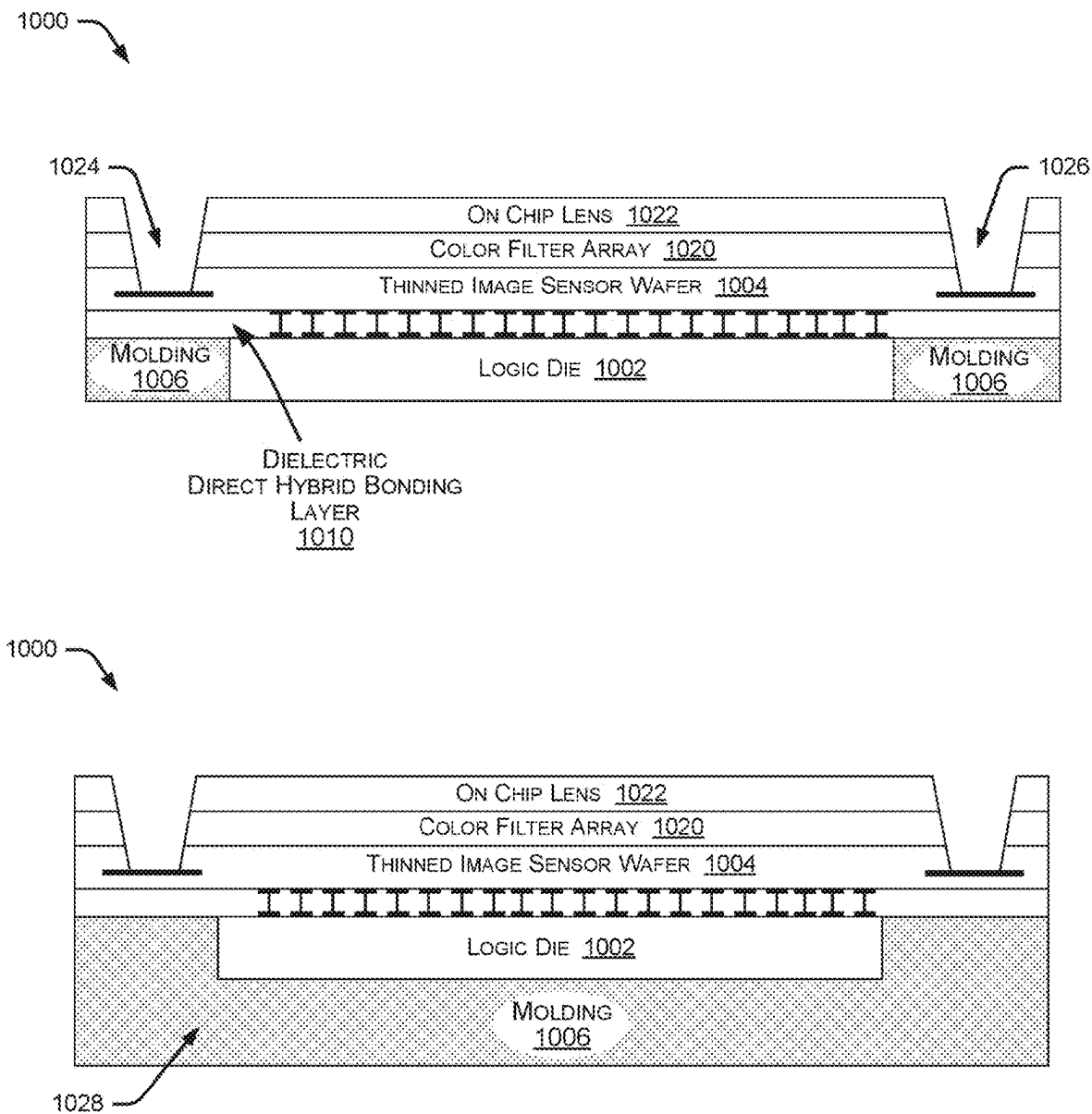
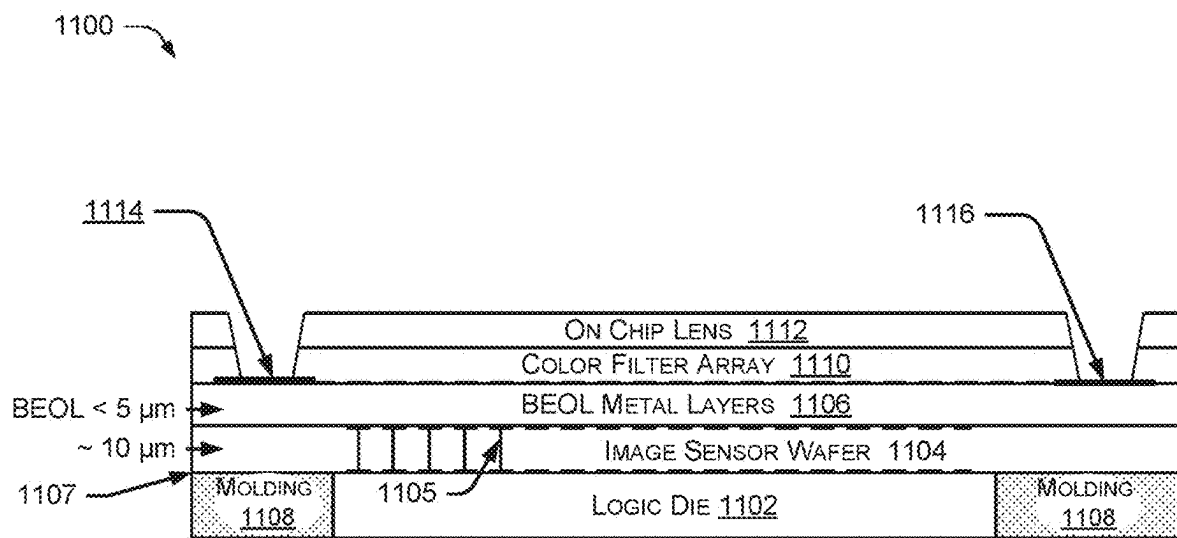


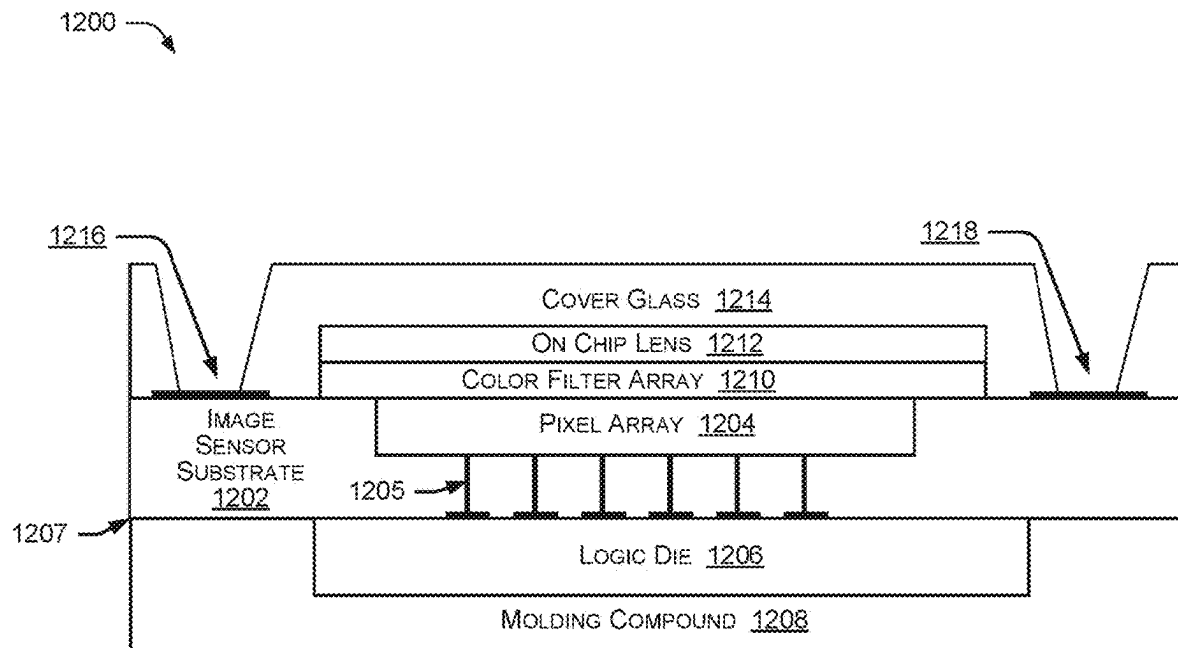
FIG. 7

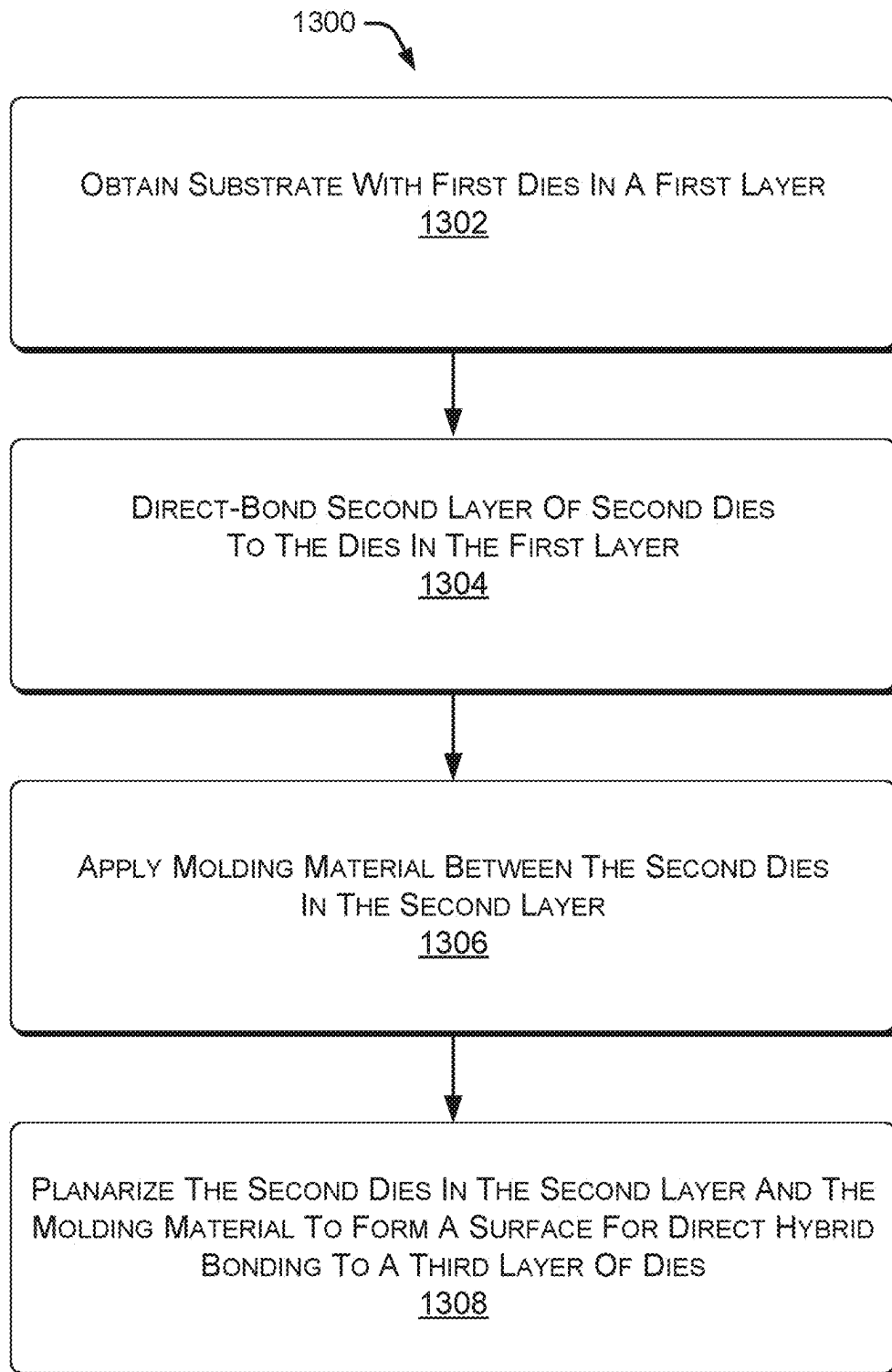
**FIG. 8**

**FIG. 9**

**FIG. 10**

**FIG. 11**

**FIG. 12**

**FIG. 13**

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STACKED DEVICES AND METHODS OF FABRICATION

PRIORITY CLAIM AND CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/586,236, filed Jan. 27, 2022, which is a continuation of U.S. patent application Ser. No. 16/413,429, filed May 15, 2019, which claims the benefit under 35 U.S.C. § 119(e)(1) of U.S. Provisional Application No. 62/671,917, filed May 15, 2018, the contents of each of which are hereby incorporated by reference in their entirety.

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

This patent application claims the benefit of priority to U.S. Provisional Patent Application No. 62/671,917 to Enquist et al., filed May 15, 2018 and incorporated by reference herein, in its entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

Conventional microelectronic packaging technologies dice wafers into individual integrated circuit microchips (“dies”) and then package the dies, or arrange the dies onto horizontal substrates and into vertical stacks that make up microelectronics packages, such as 3-dimensional ICs (“3DICs”). Separately, the fabrication processes that form the IC dies on semiconductor wafers in the first place use fabrication technologies in foundry environments that differ from the packaging environments and associated packaging techniques used in micropackaging houses. Micropackaging design is a somewhat different science than wafer fabrication design, each using respective processes and materials. Conventional wafer level packaging (WLP) creates some packaging of the dies while still on the wafer, before dicing into individual micropackages, while conventional fan-out wafer level packaging (FOWLP) places a layer of known-good-dies on a carrier wafer or panel and creates a fan-out area around the dies with molding material. A redistribution layer then fans the leads of the dies out onto the larger footprint of the molding material around the dies in the single layer, and may array connective solder balls on the larger footprint.

Both fabrication of dies and micropackaging of the fabricated dies into 3DICs and interposers seek to be more efficient and cost effective, while providing more electronic circuits and more computing power in smaller and smaller packages. Constructing a microelectronics package can adopt novel structures not constrained by the usual planarity of semiconductor fabrication on horizontal surfaces of wafers. Thus, micropackaging often uses a wider variety of materials in the construction of 3DICs and interposers, such as plastic encapsulants, polymers, and structural substrate materials that are not limited to silicon and other semiconductors. Despite their common theme of making and using IC dies, the practices of conventional semiconductor fabrication and conventional microelectronics packaging are traditionally distinct and separate, albeit related, technologies.

BRIEF DESCRIPTION OF THE DRAWINGS

Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings,

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wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various technologies described herein.

FIG. 1 is a diagram showing the start of a process for direct-bonding dies of a first footprint size to dies of a second footprint size in a wafer level process.

FIG. 2 is a diagram showing a continuation of the process of FIG. 1, for direct-bonding dies of a first footprint size to dies of a second footprint size in a wafer level process.

FIG. 3 is a diagram of example devices showing routing and interconnect variations in devices made by direct-bonding dies of a first footprint size to dies of a second footprint size in a wafer level process.

FIG. 4 is a diagram of an example micropackage and molding encapsulation fabricated by direct-bonding dies of a first footprint size to dies of a second footprint size in a wafer level process.

FIG. 5 is a diagram of a process for producing multilayered structures of stacked dies, wherein variegated dies of the various layers have various sizes, types, and dimensions.

FIG. 6 is a diagram of example interposers with variegated dies in multiple layers on one or both sides of the interposers by direct-bonding dies of a first footprint size to dies of a second footprint size in a wafer level process.

FIG. 7 is a diagram of a first example backside illuminated image sensor structure.

FIG. 8 is a diagram of a second example backside illuminated image sensor structure.

FIG. 9 is a diagram of a third example backside illuminated image sensor structure.

FIG. 10 is a diagram of a fourth example backside illuminated image sensor structure.

FIG. 11 is a diagram of a first example front-side illuminated image sensor structure.

FIG. 12 is a diagram of a second example front-side illuminated image sensor structure.

FIG. 13 is a flow diagram of an example method of fabricating direct-bonded stacked die structures in a wafer level process.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

Overview

This disclosure describes stacked devices and associated methods of fabrication. Systems and methods described herein enable die-to-wafer (D2W) or chip-to-wafer (C2W) techniques to bond layers of dies of various physical sizes, form factors, and foundry nodes (process nodes, technology nodes) to a semiconductor wafer, to interposers, or to boards and panels (all of these may be referred to herein as “substrates”). The example systems and methods enable 3D stacking of variegated dies by direct hybrid bonding within a wafer level packaging flow (WLP), including fan-out versions of 3D die stacks consisting of dies with different footprints—at the level of wafer level packaging. The systems and methods enable direct-bonding of layers of variegated dies mixed and matched from various different foundry nodes (process nodes, technology nodes), different form factors, different wafer sizes, and different package physical sizes into vertical stacks within a single micropackage at wafer level, with the benefits of high interconnect density and space-saving provided by the direct hybrid bonding. If the micropackage is an interposer, the interposer can have the variegated dies on one or both sides, including

dies of different geometries stacked vertically on one or both sides. Interposers bearing multiple types of dies of various physical sizes and foundry nodes, including stacks of such dies, can then be bonded to a semiconductor wafer, in one example scenario.

Conventional wafer-to-wafer bonding usually implies that chip dimensions are identical for the vertical wafer layers being bonded, but the yield decreases dramatically with the number of layers because individually defective dies on the wafers being stacked make an entire individual vertical stack defective. The example systems and methods described herein allow mixing and matching of variegated dies from different wafer sizes and different foundry technologies to be picked and placed on a substrate (such as a wafer), and further allows these variegated dies of different physical dimensions to form stacked layers, all through direct-bonding techniques that provide high-density signal connections and smaller packages.

These example systems and methods enable creation of micropackage architecture in which conventionally incompatible dies picked from wafers that differ from each other in technology, foundry node, wafer size, and form factor can be bonded together in the same micropackage at wafer level without much restriction, given a common interconnect interface between any two units being direct-bonded together. The example systems and methods also provide the advantage of high yield production, since variegated dies selected from different sources for direct-bonding and stacking to a wafer or interposer can be known-good-dies (KGDs).

A foundry node (“technology node,” “process node,” “process technology,” or simply “node”) refers to a specific semiconductor manufacturing process and its design rules, and generally refers to a minimum feature size. Different nodes imply different architectures and circuits from different generations of the technology. Essentially, technology node refers to the physical size of the transistor being fabricated. The more recent the technology node, the smaller the transistors, with more transistors per unit area, which in turn provides faster switching, less energy use, and cooler operation than a previous version of the same technology node. Thus, smaller technology nodes generally have smaller feature sizes: smaller transistors that are faster and more power-efficient than those made according to an earlier node.

Silicon wafer diameter nodes, a different designation than foundry nodes, have progressed in the other direction, from smaller to larger. Recent silicon wafer diameter nodes are 150 mm, 200 mm, and 300 mm, with 450 mm on the horizon. Larger semiconductor wafers reduce the manufacturing cost per transistor by raising the integration level through device scaling and other factors such as improvement in production yield because of non-defect rate, and increased throughput in production volume per unit of time. As the area of wafer surface increases, the number of semiconductor dies obtainable from a wafer also increases, reducing the production cost per die. Larger, more recent wafer sizes tend to incorporate the latest technology nodes (smaller feature sizes). Between different possible semiconductor wafer size nodes, and different foundry nodes (“technology nodes” or “process nodes”), the physical sizes and electronic feature sizes of the resulting dies can vary greatly. These different dies from different size semiconductor wafers and made according to different foundry nodes are referred to herein as “variegated dies,” that is, dies of different physical dimensions and different feature and technology sizes.

In an implementation, an example process described herein direct-bonds the variegated dies of various physical sizes, form factors, and/or foundry nodes to wafers of same or different size origin or foundry node, filling in lateral spaces caused by differences in physical sizes and form factors with a molding material to fill and complete a given horizontal bonding layer. This can be accomplished at wafer level. The molding material may be a molding compound, a resist material, a glass, a silicon spacer material, or a thermally conductive compound. The interstitial molding material may be extremely thin vertically, matching the layer’s vertical thickness determined by the heights of the dies being direct-bonded in that layer. The particular bonding layer of dies and molding may then be planarized with chemical-mechanical planarization (CMP) or other polishing technique at wafer level to prepare for a next bonding layer of variegated dies from various foundry nodes and wafer sizes to be direct-bonded to the previous layer. The example process allows continued stacking of the variegated dies and interstitial molding within each bonding layer, to make vertical stacks of the variegated dies in a microelectronics package.

Conductive through-vias, interconnects, and/or leads can be disposed in the molding material disposed laterally between the variegated dies of a given direct-bonding layer of dies. This allows horizontal fan-out of the leads of dies of a given layer, and also allows vertical transmission of signals between adjacent layers of dies, or through multiple layers of dies, without necessarily having to create any vias through the dies themselves.

The coefficient of thermal expansion (CTE) of the molding material used in the example systems and methods aims to match the CTE of silicon or other semiconductor substrate material in use. This match of the CTEs between semiconductor and molding makes the molding material amenable to expansion and contraction of the semiconductor material caused by thermal fluctuations during further steps of the fabrication process, and also amenable to thermal fluctuations during operation of the micropackage when in use as a “chip” in an end device. Likewise, the hardness, density, friability, and other physical characteristics of the molding material is ideally formulated to match those of silicon or other semiconductor, so that CMP and other polishing and finishing processes cannot distinguish between the molding material and the semiconductor, for purposes of polishing to a flat or ultra-flat surface that is free from significant dishing and rounding; and so immediately available for further direct-bonding steps in construction of vertical stacks of the variegated dies.

In an implementation, since “direct-bonding” is the only bonding process used between dies, between dies and wafer, and between dies and interposer, then intervening connection structures such as ball grid arrays may be eliminated to save space. In an implementation, direct-bonding between dies, or between dies and wafer, can be between native interconnects of the logic circuits and other native circuits of the dies and wafers. This use of native interconnects direct-bonded together can save even more space, since in some cases standard interfaces may be eliminated during design of the variegated dies themselves.

Suitable direct-bonding techniques prepare surfaces for molecule-to-molecule, covalent, ionic, and/or metallic bonding between like materials on each side of the direct-bonding interface. For example, freshly prepared flat surfaces of copper metal (Cu) may bond to each other upon contact with or without pressure and heat, forming copper diffusion bonds, metal-to-metal bonds, and the like.

Zibond® brand direct-bonding is one example direct-bonding technique for dielectrics, oxide materials, and other nonmetals (Xperi Corporation, San Jose, CA). Zibond® direct-bonding is a room temperature wafer-to-wafer or die-to-wafer direct-bonding technique. DBI® brand hybrid bonding, is an example direct-bonding technique for joining nonmetal (e.g., dielectric) surfaces that also have metal interconnects to be joined together in the same planar bonding interface (Xperi Corporation, San Jose, CA). The DBI direct hybrid bonding may be accomplished in one operation, in which oxide dielectric surfaces direct-bond together first, at room temperature, and then metal interconnects on each side of the bonding plane direct-bond together during an annealing phase at slightly or moderately elevated temperatures.

The molding material introduced above, when used as a filler or plastic encapsulant, is conventionally found in micropackaging technologies, and is also used for making reconstituted wafers having a layer of chips or dies diced from a first wafer, and then joined to a second carrier wafer or panel in a single layer on the wafer or panel. In micropackaging, such molding materials and various fillers may be put to use in layers of a few mils (a mil is one-thousandth of an inch or 0.001 inch), and up to far greater thicknesses in construction of a 3DIC.

The molding material suitable for the example systems and processes described herein is applied to fill-in lateral spaces between dies but in much thinner vertical layers than used for reconstituting wafers, at a vertical height suitable for the dies in the layer being direct-bonded to a previous layer. This molding material may be only a few microns thick (25.4 microns=1 mil), depending on the dimensions and form factor of the dies being direct-bonded to wafers and to other dies. The molding material is recruited to the wafer fabrication environment to make ultra-flat surfaces for direct hybrid bonding between layers of variegated dies. Whereas a conventional process for making a reconstituted wafer is limited to using the molding material as an encapsulant, the example processes herein are able to use the molding material to make surfaces suitable for direct-bonding stacks of variegated dies within the scope of the wafer level semiconductor fabrication process itself.

Example systems and methods are not limited to front-to-front direct hybrid bonding of dies to wafers, and front-to-front direct hybrid bonding of dies to other dies. Although front-to-front examples of stacked structures may be shown and described representatively herein, the direct bonding of dies to wafers or dies to dies according to the example system and methods described herein may be front-to-front, front-to-back, or back-to-back, when these possible configurations are desired in a given scenario. The use of handle wafers with intermediate temporary bond and debond steps may be used to present a back surface for bonding.

Example Stacked Structures

In FIG. 1 and in all the figures described herein, the layers illustrated are not shown to relative scale. For example, layers of dielectric material and silicon that provide bonding layers and direct-bonding surfaces are shown with exaggerated thickness for the sake of illustration and description, and to emphasize their presence and their various features. Some of these layers of materials may be extremely thin layers, coatings, or deposits in actual microelectronic devices.

In FIG. 1 and all the figures described herein, the various dies may be shown as direct-bonded to each other face-to-face, with direct hybrid bonding of both dielectric surfaces and metal interconnects between the dies at the bonding

interfaces but the stacked structures shown and the related processes are not limited to face-to-face direct bonding of the dies. The dies on adjacent vertical layers of the example stacked structures may also be direct-bonded face-to-back and back-to-back in addition to the face-to-face direct bonding of the dies in the layers of the stacked structures.

FIG. 1 shows the start of an example process for making a microelectronic device using a wafer level process, including die-to-wafer (D2 W) techniques. A substrate, such as a wafer, has first dies **102** in a first layer **104**, and at least some of the first dies **102** have a first footprint size and specific physical dimensions. Second dies **106** in a second layer **108** include dies with a second footprint size and/or different physical dimensions than the footprint size and physical dimensions of the first dies **102** in the first layer **104**.

Dies **102** in the first layer **104** have a front side layer of metal **110** providing electrical contacts and/or redistribution traces for integrated circuits of the dies **102**. The metal contacts **110** can be built up into a bonding layer with interconnects for direct-bonding to the second dies **106**, or in some cases the metal contacts **110** can be bonded directly to the second dies **106** without further buildup of redistribution layers (RDL) or further vertical extension using through-vias.

Likewise, the second dies **106** residing in the second layer **108** have a respective layer of metal **112** for making electrical contacts that can be built up through redistribution layers (RDL) or by vertical extension by through-vias, or in some cases direct-bonded directly to the first dies **102**.

In one implementation, dielectric (or silicon) layers **114** & **116** are added to respective dies **102** & **106** to make direct hybrid bonding surfaces. In the through-vias (e.g., **118** & **120**) or other interconnects or redistribution conductors rise vertically through the dielectric layers **114** & **116** to metal bonding pads **122** & **124** on the surfaces of the respective dies **102** & **106** for eventual direct-bonding at a bonding interface **126**.

In FIG. 2, continuing the example process, the dielectric surfaces **114** & **116** and the respective metal bonding pads **122** & **124** all participate in direct hybrid bonding at the bonding interface **126**. The two respective dielectric surfaces **114** & **116** direct-bond to each other in dielectric-to-dielectric direct bonding, while the metal bonding pads **122** & **124** direct-bond to each other in metal-to-metal direct bonding, during an annealing phase of the same direct hybrid bonding operation, for example.

The first dies **102** of the wafer, or on a substrate, may be memory dies, sensor image dies, or other kinds of dies of various types and sizes. The second dies **106**, with different physical dimensions and/or different footprints than the first dies **102**, may be logic dies or other dies to be communicatively coupled with the first dies **102**. Thus, the first dies **102** and the second dies **106** may be mixed and matched from different wafer sizes, different foundry nodes (process nodes, technology nodes), different footprints, different functionalities, different physical sizes, and so forth, referred to hereinafter as variegated dies.

In a next step, a molding material **200** is applied to at least partially fill-in spaces that occur horizontally between the second dies **106** in the second layer **108**. The molding material **200** ideally has a similar coefficient of thermal expansion (CTE) as the dies **106** or dielectric **116** (or silicon) that the molding material **200** is intervening between. The molding material may be a compound such as a filler material, a resist compound, silicon, a high thermal conductivity compound, formed diamond, deposited diamond, formed aluminum nitride, deposited aluminum

nitride, a material with a coefficient of thermal expansion (CTE) close to a CTE of silicon, a packaging material, or a board material, for example.

At this point in the example process, the wafer or substrate on which the first dies **102** reside may be thinned **202**, especially if the second layer of dies **106** is the only layer to be direct-bonded to the layer of first dies **102**.

FIG. **3** shows various ways in which first dies **102** of a wafer, carrier, or substrate that have been direct-bonded to second dies **106** of different size, footprint, wafer-origin, and/or foundry node can be interconnected externally, outside their own micropackage.

In example micropackage **300**, a metallization layer **302** of the first dies **102** has pad areas available that can be etched out of each die **102** (or wafer or substrate) to make electrical contacts, either for external connection or for continuance of the fabrication of the micropackage **300**.

In example micropackage **304**, through silicon vias (TSVs) **306** are made through the first dies **102** to enable backside electrical connection at backside contact pads **308**, for example, and/or construction of redistribution layers (RDLs) on the backside of the dies **102** or thinned wafer hosting the dies **102**.

In example micropackage **310**, through-silicon vias (TSVs) **312** are made through the second dies **106** to enable electrical connection at backside contact pads **314** of the second dies **106**, for example, and/or to connect with redistribution layers (RDLs) to be fabricated on the backsides of the second dies **106**.

Through-vias **316** may also be constructed to pass through the molding material **200** that has been applied to at least partially fill the spaces between dies **106** of a given direct-bonded layer. Such through vias **316** & **318** through the molding material **200** can extend to either side of the entire micropackage **310**. Through-vias **320** can also extend from the contact pads **122** of the first dies **100**, through the molding material **200**, to the backside of the second dies **106** or beyond (if there are additional direct-bonded layers of dies). Similarly, through-vias **322** can extend all the way through the entire example micropackage **310**, including through the molding material **200**, from one side of the example micropackage **310** to the other.

In example micropackage **324**, various through-silicon vias (TSVs) **306** & **312** and various through-vias **316** extending through the molding material **200** can extend from the contact pads of dies **102** & **106** from multiple layers of the micropackage **324**, to both front and back sides of the microelectronics package **324**, traversing through the molding material **200** as needed.

FIG. **4** shows another example microelectronics package **400** with molding material between dies also encapsulating the dies or filling out package dimensions. First dies **402** of a first type, first footprint, and first dimensions are direct-bonded to dies **404** of a second type, second footprint, and second dimensions, with intervening dielectric bonding layers **114** & **116**. The dies **402** & **404** are physically and electrically connected by direct hybrid bonding, in which the dielectric layers **114** & **116** direct-bond to each other with dielectric-to-dielectric direct bonds, and metal interconnects, pads, and/or vias direct-bond to each other across the bonding interface with metal-to-metal direct bonds.

In this embodiment, a molding material **406** is applied to at least partially fill-in spaces that occur horizontally between the second dies **404**. The molding material **406** also forms a layer **408** above the top of the second dies **404**, to encapsulate the dies **404**, complete the package, fill a void, and/or to form a filler layer **408** for continuing fabrication of

the micropackage **400** above the top of the layer **408** of the molding material **406**. The molding material **406** may also fill-in vertical spaces above some dies **404** of the second layer that are shorter in vertical height than other dies **404** of the second layer. Conductive leads of the shorter second dies **404** may be extended with through-vias to a top surface of the second layer of dies **404**, or to a next layer of dies, or to an outside surface of the example micropackage **400**.

The molding material **406** ideally has a similar coefficient of thermal expansion (CTU) as the dies **404** or dielectric **116** (or silicon) that the molding material **406** is intervening between and/or encapsulating. The molding material **406** may be a compound such as a filler material, a resist compound, silicon, a high thermal conductivity compound, formed diamond, deposited diamond, formed aluminum nitride, deposited aluminum nitride, a material with a coefficient of thermal expansion (CTE) close to a CTE of silicon, a packaging material, or a board material, for example.

Through vias **410** & **412** & **414** & **416** can extend from any contact pads of any of the dies **402** & **404** in the micropackage **400** to any side of the micropackage **400** for external connection, traversing through the molding material **406** as needed.

The first dies **402** and/or the molding material **406** may be thinned as needed for further layering or packaging.

FIG. **5** show an example process for making a microelectronic device **500** with variegated dies direct-bonded in multiple layers. The variegated dies of FIG. **5** can be logic dies, image sensor dies, high bandwidth memory (HBM) dies, and so forth.

In FIG. **5**, all direct bonds or direct hybrid bonds shown between dies can be direct bonds formed between surfaces of the dies themselves, or can be direct bonds formed via one or more intervening dielectric bonding layers (not shown explicitly). The dielectric bonding layers (not shown in FIG. **5**) can contain metal interconnects or through-vias for vertically extending electrical contacts through the dielectric bonding layers. Examples of dielectric bonding layers **114** & **116** can be seen in FIGS. **1-2**.

A substrate, carrier or wafer **100** has first dies **102**, and at least some of the first dies **102** have a first footprint size and specific physical dimensions. Second dies **106** with a second footprint size and/or different physical dimensions than the footprint size and physical dimensions of the first dies **102**.

Contact pads of the dies, electrical connections, interconnects, fan-out lines, redistribution traces, and redistribution layers (RDLs) are not shown explicitly in FIG. **5**, but leads of the dies **102** & **106** are direct-bonded together across the bonding interfaces either directly, or through vertical interconnects in dielectric bonding layers between the dies **102** & **106**.

Molding material is placed horizontally between the dies **106** to at least partly fill the empty spaces between the dies **106**.

The top surface of the second dies **106** and the molding material can be planarized to make a surface for direct bonding or direct hybrid bonding another layer of variegated dies on top of the second dies **106**. The planarization can be chemical mechanical polishing (CMP) to impart flatness and surface characteristics sufficient for direct bonding.

Third dies **500** can be direct bonded or direct hybrid bonded to the second dies **106**. The third dies **500** can be variegated dies with footprints, foundry nodes, wafer origins, functionalities, and physical dimensions different than second dies **106**, and may also be different than the first dies **102**. The various dies **102** & **106** & **500** do not have to be different from each other in size, footprint, foundry node,

functionality, and so forth, but the example process shown in FIG. 5 makes such stacking of variegated dies 102 & 106 & 500 possible in a wafer level process.

Another round of molding material, which may be the same material or a different material than molding material used in the second layer, is applied between the third dies 500 in the third layer to at least partly fill the empty spaces horizontally between the dies 500 of the third layer. The top surfaces of the third dies 500 and the molding material of the third layer are planarized to make a surface for direct bonding or direct hybrid bonding another layer of variegated dies, if desired, on top of the third layer of dies 500. The planarization can be chemical mechanical polishing (CMP) to impart flatness and surface characteristics sufficient for direct bonding.

The process can continue with direct-bonding or direct hybrid bonding of each new layer of variegated dies, filling-in intervening spaces between dies of each new layer with molding material, and then planarizing the top of the dies and the molding material for direct bonding of the next layer of dies.

Dies 102 in the first layer 104 have a front side layer of metal 110 providing electrical contacts and/or redistribution traces for integrated circuits of the dies 102. The metal contacts 110 can be built up into a bonding layer with interconnects for direct-bonding to the second dies 106, or in some cases the metal contacts 110 can be bonded directly to the second dies 106 without further buildup of redistribution layers (RDL) or further vertical extension using through-vias.

Although not shown in FIG. 5, conductive through-vias, conductive interconnects, leads, redistribution traces, fan-outs, and redistribution layer (RDLs) can be routed in the microelectronics package, including through the molding materials 510 & 518, to connect the dies 502 & 506 & 514 to each other, and/or to contact pads on the outside of the microelectronics package.

FIG. 6 shows different example configurations of interposer devices built according to the processes described herein. The example interposers can be used in 2.5 D integrated circuit technology, for example, to provide several benefits.

First, variegated dies of various different footprint sizes, types, foundry nodes, wafer origins, physical dimensions, and so forth, can be mixed and matched not only on one side, but on both sides of an interposer substrate that is silicon, organic, or other material. The included variegated dies can be bonded together into the interposer package with direct bonding techniques instead of solder ball arrays, for example, although solder ball arrays are not excluded. The direct bonding or direct hybrid bonding can create ultra-high density routing between dies. Moreover, KGDs (known good dies) can be used to construct the example interposers, increasing production yield.

The interposers can then be reconstituted onto a semiconductor wafer, or other silicon or organic substrate, for example, or onto the package substrate of microelectronic devices making an efficient way to create such micropackages using wafer level processes to provide smaller size and higher bandwidth, and with less expensive techniques than are possible with conventional ball grid array interconnects and conventional wafer reconstitution processes.

The example interposers shown and described can help to decrease interconnection length between multiple dies assembled on the example interposers. This increases the number of interconnection routes for the interposer with

stacked structure of variegated dies, saves power consumption, decreases latency, and increases bandwidth compared to conventional interposers.

A first example interposer 601 has an interposer substrate 600 with a first layer of dies 602 and a second layer of different dies 604 direct-bonded to the first layer of dies 602. A molding material 606 at least partly fills-in the spaces between the dies 604 in the second layer. The molding material 606 may also continue horizontally over the dies 604 to encapsulate the dies 604 (not shown) and complete the interposer package. The interposer 600 may use one or more redistribution layers (RDLs) 608 to distribute or fan-out traces. Interconnects and through-vias 610 can be leveraged anywhere in the package, including through the molding material 606 depending on application and need (not shown). The example through-vias 610 can also pass through the interposer substrate 600 and connect to a package substrate, to a wafer, or even to another interposer via contact pads disposed on an external surface of the interposer substrate 600, for example (not shown).

A second example interposer 611 with an interposer substrate 612 has a layer of first dies 614 of a first type, and a layer of second dies 616 of a second type direct-bonded to the first dies 614. The dies 614 & 616 may differ from each other in footprint size, functionalities, physical size, foundry node, wafer origin, and in many other specifications, but can be combined together in a wafer level process using the molding material 618 to make direct-bondable layers for making multiple levels of stacked dies 614 & 616 . . . n, in the example interposer. The interposer 611 may interface with a wafer or other devices via its interposer substrate 612 or via die 616 positioned to present front contacts of backside TSV reveals to the interface (not shown).

A third example interposer 621 has an interposer substrate 620, a layer of first dies 622 of a first type on distribution lines 623 or on one or more RDLs, for example, and a second layer of dies 624 of a second type direct-bonded to the first layer of dies 622. The dies 622 & 624 do not have to be different types, they can be the same type of die, but the example process and structures allows the dies 622 & 624 to be very different from each other. A molding material 625 fills-in spaces between dies 624 of the second layer and can also encapsulate (not shown) the dies 624 or can encapsulate (not shown) the entire top side of the interposer package 621.

On an opposing side of the interposer substrate 620, a third layer of dies 626 are bonded and electrically connected if applicable, with another molding material 627 or the same molding material used above filling in spaces between dies 626. A fourth layer of dies 628, which is the second layer on this opposing side of the interposer substrate 620 are direct-bonded to the third layer of dies 626, with another molding material 629 filling-in spaces between the dies 628 of the fourth layer and can also encapsulate (not shown) the dies 628 or can encapsulate (not shown) the entire bottom side of the interposer package 621. Through-vias, distribution traces, and contact pads can be disposed anywhere in, or on the outside of the interposer package 621, traversing any of the molding material 625 & 627 & 629 as needed for routing. Thus, the interposer package 621 can interface with a wafer or other devices via either top or bottom connections, or both.

The example interposers 601 & 611 & 621 can be used to make high bandwidth memories (HBM), with high density routing. Memory dies can be stacked on the interposers along with dies providing the main processor and HBM controller. The example interposers 601 & 611 & 621 can

also be used for other high performance computing applications, artificial intelligence (AI), graphic processors, image sensors, and other applications.

Example Image Sensor Structures and Processes

Example image sensor devices are now described as example embodiments of the stacked die structures and associated processes shown and described above in FIGS. 1-6. Both backside illuminated and front-side illuminated image sensors and associated methods of fabrication are described using the processes of making stacked die structures of FIGS. 1-6.

Front-side illumination refers to light entering photo diode elements through a (BEOL) layer of metal wiring built on the front side of the photo diode elements. A bond interface with a logic chip is on the backside of the photo diode elements die, faced away from the incoming light. Back-side illumination refers to light entering the photo diode elements on the side of the backside of the photo diode elements die, with the BEOL metal wiring layer on the opposing front side of the photo elements die. The BEOL metal wiring layer (on the front side, away from the incoming light) faces the bond interface with the logic chip.

This disclosure is not limited to the use of image sensor wafers and is more generally applicable to semiconductor wafers, for example CMOS wafers, which have a BEOL front side disposed on an active device layer which can be facing toward or away from a bond interface.

FIG. 7 is a diagram of a first example backside illuminated image sensor structure 700. Generally, the example backside illuminated image sensor structure 700 can be made by direct hybrid bonding of logic dies 702 of a first footprint size, to a front side of an image sensor wafer 704 in a die-to-wafer (D2 W) process, and optionally filling-in spaces between the logic dies 702, either fully or partly, with a molding compound 706, a resist material, a silicon spacer material, glass, or a thermally conductive compound, for example. The direct hybrid bonding between photo diode element dies on the image sensor wafer 704 and the logic dies 702 may be intermediated by dielectric layers 708 & 710 with through-vias 712 & 714 and contact pads 716 & 718 suitable for direct hybrid bonding.

The image sensor wafer 704 may be a 200 millimeter wafer, or of other size, and the logic dies 702 can also be of various sizes, for example logic dies 702 from a 300 millimeter wafer, for example.

The direct hybrid bonding can be a direct bond interconnect (DBI®) hybrid bonding process, for example, or other process that direct-bonds dielectric surfaces together while direct-bonding the metal contact pads 716 & 718 together in steps of the same direct-bonding operation.

A support carrier (not shown in FIG. 7) may optionally be direct bonded to at least the logic dies 702 using an intervening dielectric, for example. Imaging layers such as a color filter array 720 and on-chip lens 722, for example, may be deposited, bonded or direct-bonded to the image sensor wafer 704.

Last metal contact areas 724 & 726 may be etched out to make package interconnects, or other interconnect methodologies may be implemented through routing lines and through-vias, including routing lines and through-vias directed through the molding material 706.

An example process for making the example backside illuminated image sensor structure 700 includes obtaining an image sensor wafer 704 that has photo-diode arrays or pixel arrays, the arrays of a first size. The example process creates microcircuits and bonding pads on a front-side of the image sensor wafer 704, the bonding pads suitable for direct hybrid

bonding. The process continues with obtaining a logic die 702 of a second size, for example, the second size can be smaller than the first size of the photo-diode arrays or the pixel arrays of the image sensor wafer 704. Microcircuits and bonding pads for direct hybrid bonding are also made on the logic die 702.

Next, the example process joins the logic die 702 to at least one photo-diode array or pixel array of the image sensor wafer 704 with a hybrid bonding technique to form a mechanical and electrical connection between the bonding pads of the logic die 702 and the bonding pads of the image sensor wafer 704.

The molding material 706 may be added to extend the logic die, which may have a variety of sizes, to match the larger size of the photo-diode array or pixel array when needed, or to partially or fully fill in horizontal spaces in the horizontal layer containing the logic die(s) 702. The molding material 706 can be a molding compound, passive silicon, a glass, a thermally conductive material, or other suitable material. The filler material may also be formed or deposited diamond, formed or deposited aluminum nitride, a material with a coefficient of thermal expansion (CTE) close to a CTE of silicon, a packaging material, and a board material, for example. The molding material 706 may have a high thermal conductivity, and may extend below the logic dies 702 to dissipate heat and hot spots in the image sensor wafer 704 (not shown). The molding material 706 may also extend below the logic dies 702 to add mechanical strength to the layer of the logic dies 702 (not shown). If passive or dummy silicon is the filler material, the silicon may be applied as a spin coating, for example.

Structure 700' shows a variation of the example backside illuminated image sensor package 700', in which access to the bonding pads 726 & 726 is obtained through vias created on the front side of the image sensor package 700'.

In the example process, the joining may direct-bond logic dies 702 of one size, from a 300 millimeter wafer, for example, to image sensors 704 on a 200 millimeter image sensor wafer 704, for example. The direct-bonding can additionally be a direct hybrid bonding process, such as a direct bond interconnect (DBI®) hybrid bonding process for fine-pitch hybrid bonding. The bonding can also be a direct oxide bonding process, such as oxide bonding in the context of TSVs, for the electrical interconnects. A suitable example oxide bonding technique for this scenario can be a ZIBond® direct oxide bonding process, for example, or another direct-bonding process. The photo-diode arrays or pixel arrays of the image sensor wafer 704 may comprise CMOS image sensors (CIS), or other types of image sensors.

The logic dies 702 may include logic driver circuitry or memory, such as DRAM, or both logic driver circuitry and DRAM memory for the photo-diode arrays or pixel arrays. The photo-diode arrays or pixel arrays are on a backside of the image sensor wafer 704 for BSI implementations, and the example process may include thinning the backside of the image sensor wafer 704.

The example process continues with depositing the color filter array 720 and an on-chip lens 722 onto the thinned backside of the image sensor wafer 704. Then, the process includes creating openings to the contact pads 724 & 726 of the photo-diode arrays or pixel arrays through a backside of the image sensor wafer 704. The contact pads 724 & 726 are usually chip-to-package bonding sites. In a variation, openings to the bonding pads 724 & 726 can also be made from the front-side 728 of the image sensor package, an in example backside illuminated image sensor structure 700'. In structure 700', the openings to the contact pads 724 & 726

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are made through the molding material **706** to the front-side **728** of the image sensor wafer **704**. In another implementation, the openings to the contact pads **724** & **726** are made through a bond via array (BVA) process.

FIG. **8** shows a second example backside illuminated image sensor structure **800** & **802**. A reconstituted wafer **800** is fabricated by bonding logic dies **804** & **806** and vertically shorter logic die **808**, for example, to a carrier wafer **809**. The logic dies **804** & **806** & **808** and carrier **809** may optionally be bonded or adhered to a back support **810** for extra strength if desired. Horizontal spaces between the logic dies **804** & **806** & **808** are at least partially filled with a molding material **812**. A direct hybrid bonding layer **816** of dielectric material and metal interconnects tops off the reconstituted logic package **800**.

In an image sensor portion **802** of the package, an image sensor wafer **814** of optionally different size and foundry origin than the logic dies **804** & **806** & **808** is surmounted by a color filter array **820**, and an on chip lens **822**. The image sensor wafer **814** is thinned, and a direct hybrid bonding layer **818** of dielectric and metal interconnects is added. The reconstituted logic part of the package **800** and the image sensor portion of the package **802** are now direct hybrid bonded together to make the backside illuminated image sensor. Bond pads of the thinned image sensor wafer **814** are exposed for connection to a microelectronic device.

An example process for making the example backside illuminated image sensor structure **800** & **802** of FIG. **8** includes creating conductive vias **807** in the carrier wafer **809**, bonding the logic dies **804** & **806** & **808** of a first footprint size, such as dies from a 300 millimeter wafer, to the conductive vias **807** of the carrier wafer **809** using direct hybrid bonding or a surface mount technology, adding the molding material **812** between the logic dies **804** & **806** & **808** to make a reconstituted logic wafer **800** on the carrier wafer **809**, thinning the carrier wafer **809** to expose the conductive vias **807** on an opposing side of the carrier wafer **809**, and adding a direct bonding layer **816** to the exposed conductive vias **807** of the carrier wafer **809**. An optional back support wafer **810** may be attached to the reconstituted logic wafer. The example process further includes adding a direct bonding layer **818** to an image sensor wafer **814**, and then joining the reconstituted logic wafer **800** to the image sensor wafer **802** by bonding the respective direct hybrid bonding layers **816** & **818** to each other.

At this point, the example process then includes thinning a backside of the image sensor wafer **814**, and depositing a color filter array **820** and an on-chip lens **822** on the thinned backside of the image sensor wafer **814**. The process then opens vias to contact pads **824** & **826** of the image sensor wafer **814** through the color filter array **820** and the on-chip lens **822**.

The molding material **812** may be a mold compound, a resist compound, silicon, a high thermal conductivity compound, formed or deposited diamond, formed or deposited aluminum nitride, a material with a coefficient of thermal expansion (CTE) close to a CTE of silicon, a packaging material, and a board material. The molding material may have a high thermal conductivity and may extend below the logic dies **804** & **806** & **808** to dissipate heat and hot spots in the thinned image sensor wafer **814** (not shown). The molding material **812** may also extend below the logic dies **804** & **806** & **808** to add a mechanical strength to the layer of the logic dies **804** & **806** & **808** (not shown).

FIG. **9** shows another example backside illuminated image sensor structure **900**. This example structure **900** attaches logic dies **902** to a thinned handle wafer **910** during

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fabrication. Handle wafer **910** may also or alternatively be thinned after logic die **902** attachment. The example process creates vias **905** through the handle wafer **910** for coupling the logic dies **902** to the thinned image sensor wafer **904**. Additionally or alternatively, the vias **905** may be formed by a via first, via middle, or via last process and some or all of the interconnections could be made by more traditional means, including through the use of solder connections.

An example process for making the example backside illuminated image sensor structure **900** of FIG. **9** includes planarizing the image sensor wafer **904** to make a thinned image sensor wafer **904** that has chip-to-package conductive contact pads **924** & **926** and has conductive bonding pads for connection to the logic dies **902**. The process direct-bonds a front-side of the thinned image sensor wafer **904** to a first handle wafer **910**, attaching a temporary second handle wafer (not shown) to a backside of the thinned image sensor wafer **904** using a temporary dielectric layer that is direct-bonded between a backside of the thinned image sensor wafer **904** and the temporary second handle wafer. Image sensor wafer **904** may also or alternatively be thinned after direct-bonding to the first handle wafer **910**. The process thins the first handle wafer **910** attached to the front-side of the thinned image sensor wafer **904**, and creates the conductive through-silicon vias (TSVs) **905** through the thinned first handle wafer **910**, with first ends of the TSVs **905** conductively coupled to the bonding pads of the image sensor wafer **904**. The process adds a layer of bonding pads to second ends of the TSVs **905**, then direct-bonds the logic dies **902** to the bonding pads on the thinned first handle wafer **910**. The process adds a molding compound **906**, resist material, or silicon spacers in the horizontal spaces around the logic dies **902**, then removes the temporary first handle wafer (not shown) and the temporary dielectric layer from the backside of the thinned image sensor wafer **904**. The process exposes the conductive contact pads **924** & **926** of the thinned image sensor wafer **904** from the backside of the thinned image sensor wafer **904**, and then deposits a color filter array **920** and an on-chip lens **922** onto the backside of the thinned image sensor wafer **904**. Conductive contact pads **924** & **926** of the thinned image sensor wafer **904** may alternatively be exposed after deposition of the color filter array **920** and/or on-chip lens **922**.

The thinned image sensor wafer **904** may be a 200 millimeter wafer, and the logic dies can be from a 300 millimeter wafer, for example.

For bonding techniques, the example process may include direct hybrid bonding the logic dies **902** to the layer of bonding pads on the thinned first handle wafer **910**. The direct hybrid bonding may be direct bonding interconnect (DBI®) brand of direct hybrid bonding, for example. The example process may include direct hybrid bonding the first ends of the TSVs **905** to the bonding pads of the image sensor wafer **904**. The temporary dielectric layer can be direct bonded between the backside of the thinned image sensor wafer **904** and the temporary second handle wafer (not shown) using a ZiBond® direct bonding process, for example.

In an implementation, the image sensor wafer **904** has a 3-10 micron thickness of silicon and less than a 5 micron thickness of back end of line (BEOL) layers. The first handle wafer **910** can either be thinned to greater than approximately 10 microns for greater downstream mechanical integrity or thinned to less than approximately 10 microns for simplifying subsequent processing.

The second handle wafer (not shown) can be a 200 millimeter wafer with an initial thickness of approximately

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750 microns. A layer containing the logic dies and the molding compound, the resist material or silicon spacers can be greater than 50 microns thick. The molding compound **906** can be a mold material, a resist compound, silicon, a high thermal conductivity compound, formed or deposited diamond, formed or deposited aluminum nitride, a material with a coefficient of thermal expansion (CTE) close to a CTE of silicon, a packaging material, and a board material. The molding compound **906** may have a high thermal conductivity and may extend below the logic dies **902** in order to dissipate heat and hot spots in the thinned image sensor wafer **904** (not shown). The molding compound **906** may also extend below the logic dies **902** to add mechanical strength to the layer of the logic dies **902** (not shown).

FIG. **10** shows another example backside illuminated image sensor structure **1000**, which features fabrication using a temporary image sensor handle wafer, without a permanently incorporated handle wafer **910** of FIG. **9**.

An example process for making the image sensor structure **1000** of FIG. **10** includes planarizing an image sensor wafer **1004** to make a thinned image sensor wafer **1004**, wherein the thinned image sensor wafer has chip-to-package conductive contact pads **1024** & **1026** and has conductive bonding pads for connection to logic dies **1002**. The process direct-bonds a front-side of the thinned image sensor wafer **1004** to a first handle wafer (not shown) via a first dielectric layer (not shown), then direct-bonds the backside of the thinned image sensor wafer **1004** to a second handle wafer (not shown) via a second dielectric layer. The process removes the first handle wafer from the front side of the thinned image sensor wafer **1004** while leaving the first dielectric layer in place, adding direct hybrid bond conductors and through dielectric vias (TDVs) **1005** to the first dielectric layer to convert the first dielectric layer to a direct hybrid bonding layer **1010**. The process bonds logic dies **1002** to the direct hybrid bonding layer **1010** via direct hybrid bonds, and adds a molding compound **1006** into the horizontal spaces around the logic dies **1002**. The molding compound **1006** can be a filler, a resist material, or silicon spacers, for example. The process then removes the second handle wafer (not shown) and the second dielectric layer from the backside of the thinned image sensor wafer **1004**, and exposes the conductive contact pads **1024** & **1026** of the thinned image sensor wafer **1004** from the backside of the thinned image sensor wafer **1004**. The process deposits a color filter array **1020** and an on-chip lens **1022** onto the backside of the thinned image sensor wafer **1004**.

The thinned image sensor wafer **1004** can be a 200 millimeter wafer, and the logic dies **1002** may be diced from a 300 millimeter wafer, for example.

The molding compound **1006** can be a filler material, a resist compound, silicon, a high thermal conductivity compound, formed or deposited diamond, formed or deposited aluminum nitride, a material with a coefficient of thermal expansion (CTE) close to a CTE of silicon, a packaging material, and a board material. The molding compound **1006** can have a high thermal conductivity and may extend below the logic dies **1002** as shown at **1028** to dissipate heat and hot spots in the thinned image sensor wafer **1004**, and/or provide mechanical strength to the layer of logic dies **1002**.

FIG. **11** shows a first example of a front-side illuminated image sensor structure **1100**, made via the example processes described with respect to FIGS. **1-6**. FIG. **11** can be distinguished from the sensor structures of FIGS. **7-10**, which are backside illuminated sensor structures. In the front-side illuminated image sensor structure **1100** of FIG.

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11, light shines through BEOL metal layers **1106** on a front side of an image sensor wafer **1104**.

An example process for fabricating the example front-side illuminated image sensor structure **1100** includes building back-end-of-line (BEOL) layers **1106** on a front side of an image sensor wafer **1104**, direct-bonding a handle wafer (not shown) onto the BEOL layers via a dielectric layer (not shown), thinning a backside of the image sensor wafer **1104** to approximately 10 microns, for example; creating through-vias **1105** through the thinned image sensor wafer **1104**, building a direct hybrid bond layer **1107** conductively coupled with ends of the TSVs **1105** on the backside of the thinned image sensor wafer **1104**, and direct hybrid bonding logic dies **1102** to the direct hybrid bonding layer **1107**.

The example process further continues with adding a filler or molding material **1108** around the logic dies **1102**. The filler material can be a mold material **1108**, a resist compound, silicon, a high thermal conductivity compound, formed or deposited diamond, formed or deposited aluminum nitride, a material with a coefficient of thermal expansion (CTE) close to a CTE of silicon, a packaging material, and a board material. The filler material **1108** may have a high thermal conductivity and may extend below the logic dies **1102** to dissipate heat and hot spots in the thinned image sensor wafer **1104** and/or provide mechanical strength to the layer of logic dies **1102** (not shown).

The example process continues with removing the handle layer (not shown) and the dielectric (not shown) layer from the (top of the) BEOL layers **1106**, opening contact pads **1114** & **1116** of the BEOL layers **1106** from a front side of the BEOL layers **1106**, and depositing a color filter array **1110** and an on-chip lens **1112** to the BEOL layers **1106** to complete a front side illuminated image sensor chip **1100** with logic dies **1102** direct hybrid bonded to the backside of the thinned image sensor wafer **1104**.

The thinned image sensor wafer **1104** can be a 200 millimeter wafer, and the logic dies can be from a 300 millimeter wafer, for example. A thickness for the logic dies **1102** can be selected that provides sufficient mechanical strength for the package **1100**.

FIG. **12** shows a second example front-side illuminated image sensor structure **1200**. This implementation of a front-side illuminated image sensor structure **1200** uses molding material **1208** to fill-in packaging around logic dies **1206** of one size, bonded to image sensor substrates **1202** and pixel arrays **1204** of different sizes, all in a wafer level production process.

An example process making the front-side illuminated image sensor structure **1200** of FIG. **12** includes bonding a color filter array **1210** and on-chip lens **1212** to a front side of a pixel array **1204** on an image sensor substrate **1202** and then adding a cover glass **1214** over the on-chip lens **1212**.

The example process then thins the backside of the image sensor substrate **1202**, and creates through-vias **1205** and a direct hybrid bonding layer **1207** on the backside of the image sensor **1202**. The process attached logic dies **1206** to the direct hybrid bonding layer **1207** on the backside of the image sensor **1202**, and adds molding compound **1208** or a high strength thermally conductive material laterally to the logic die **1206** and to the lateral edges of the image sensor **1202**. The molding compound **1208** or the high strength thermally conductive material can extend in a layer below the logic dies **1206**.

The thinned image sensor **1202** can be from a 200 millimeter image sensor wafer, and the logic dies **1206** can

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be from a 300 millimeter wafer, for example. Contact pads **1216 & 1218** for package bonding are opened from the front side of the device.

Example Methods

FIG. **13** shows an example method **1300** of fabricating direct-bonded stacked die structures in a wafer level process. Operations of the example method **1300** are shown in individual blocks.

At block **1302**, a substrate with first dies in a first layer is obtained.

At block **1304**, a second layer of second dies is direct-bonded to the first dies in the first layer.

At block **1306**, a molding material is applied between the second dies in the second layer.

At block **1308**, the second dies in the second layer, and the molding material, are planarized to form a surface suitable for direct hybrid bonding to a third layer of dies.

In the foregoing description and in the accompanying drawings, specific terminology and drawing symbols have been set forth to provide a thorough understanding of the disclosed embodiments. In some instances, the terminology and symbols may imply specific details that are not required to practice those embodiments. For example, any of the specific dimensions, quantities, material types, fabrication steps and the like can be different from those described above in alternative embodiments. The term “coupled” is used herein to express a direct connection as well as a connection through one or more intervening circuits or structures. The terms “example,” “embodiment,” and “implementation” are used to express an example, not a preference or requirement. Also, the terms “may” and “can” are used interchangeably to denote optional (permissible) subject matter. The absence of either term should not be construed as meaning that a given feature or technique is required.

Various modifications and changes can be made to the embodiments presented herein without departing from the broader spirit and scope of the disclosure. For example, features or aspects of any of the embodiments can be applied in combination with any other of the embodiments or in place of counterpart features or aspects thereof. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense.

While the present disclosure has been disclosed with respect to a limited number of embodiments, those skilled in the art, having the benefit of this disclosure, will appreciate numerous modifications and variations possible given the description. It is intended that the appended claims cover such modifications and variations as fall within the true spirit and scope of the disclosure.

What is claimed is:

1. A microelectronic device comprising:

a first die layer comprising:

a first die having a first lateral footprint and a plurality of first through silicon vias (TSVs) extending vertically at least partially through the first die;
a first dielectric layer on the first die;
a plurality of first metal bonding pads at least partially embedded in the first dielectric layer over and electrically connected to the plurality of first TSVs; and
a first encapsulant disposed over a side surface of the first die; and

a second die layer comprising:

a second die having a second lateral footprint different from the first lateral footprint;
a second dielectric layer on the second die;

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a plurality of second metal bonding pads at least partially embedded in the second dielectric layer; and

a second encapsulant disposed over a side surface of the second die,

wherein the first dielectric layer is directly bonded to the second dielectric layer, wherein the plurality of first metal bonding pads is directly bonded to the plurality of second metal bonding pads, and wherein at least a portion of at least one second metal bonding pad is disposed directly over at least one first TSV.

2. The microelectronic device of claim 1, wherein the first encapsulant comprises glass.

3. The microelectronic device of claim 2, wherein the second encapsulant comprises glass.

4. The microelectronic device of claim 1, wherein at least one of the first encapsulant and the second encapsulant comprises a silicon-containing material.

5. The microelectronic device of claim 1, wherein at least one of the first encapsulant and the second encapsulant comprises a molding material.

6. The microelectronic device of claim 1, wherein at least one of the first encapsulant and the second encapsulant has a coefficient of thermal expansion (CTE) close to a CTE of silicon.

7. The microelectronic device of claim 1, wherein the second die layer comprises a plurality of additional dies hybrid bonded to the first die layer.

8. The microelectronic device of claim 7, wherein the second encapsulant is disposed between the second die and a third die of the plurality of additional dies.

9. The microelectronic device of claim 1, wherein the first die and the second die have different functionalities.

10. The microelectronic device of claim 1, wherein the second die comprises a memory.

11. The microelectronic device of claim 1, further comprising a substrate directly bonded to the second die layer.

12. The microelectronic device of claim 1, wherein the first and second die layers are connected without a redistribution layer (RDL) therebetween.

13. The microelectronic device of claim 1, wherein at least one of the first encapsulant or the second encapsulant comprises multiple layers.

14. A microelectronic device comprising:

a first die having a first lateral footprint and a first through silicon via (TSV) extending vertically at least partially through the first die, a first dielectric layer disposed on the first die with a first metal bonding pad at least partially embedded in the first dielectric layer connected with the first TSV;

a second die having a second lateral footprint different from the first footprint, a second dielectric layer disposed on the second die with a second metal bonding pad at least partially embedded in the second dielectric layer, the first dielectric layer directly bonded to the second dielectric layer and the first metal bonding pad directly bonded to the second metal bonding pad,

at least one additional die directly bonded to the first dielectric layer; and

an encapsulant disposed over a side surface of the second die,

wherein the first metal bonding pad and the second metal bonding pad provide vertical electrical transmission between the first and second dies, and wherein at least a portion of the second metal bonding pad is disposed directly over the first TSV.

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15. The microelectronic device of claim 14, wherein the encapsulant comprises glass.

16. The microelectronic device of claim 14, further comprising a second encapsulant along a side surface of the first die.

17. The microelectronic device of claim 14, further comprising a plurality of additional dies hybrid bonded to the second dielectric layer and corresponding pads in the second dielectric layer.

18. The microelectronic device of claim 14, wherein the at least one additional die is hybrid bonded to the first dielectric layer and corresponding pads in the first dielectric layer.

19. The microelectronic device of claim 14, further comprising a substrate directly bonded to the second die.

20. The microelectronic device of claim 14, wherein the first and second dies are connected without a redistribution layer (RDL) therebetween.

21. The microelectronic device of claim 14, wherein the first metal bonding pad and the second metal bonding pad provide vertical electrical transmission of a signal between the first and second dies.

22. The microelectronic device of claim 14, wherein the first metal bonding pad connected with the first TSV comprises one or more of an electrical connection and a physical connection between the first metal bonding pad and the first TSV.

23. The microelectronic device of claim 14, wherein the encapsulant is present in a region laterally adjacent to the side surface of the second die.

24. The microelectronic device of claim 14, wherein the encapsulant directly contacts the side surface of the second die.

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25. A microelectronic device comprising:

a first die having a first lateral footprint and a first through silicon via (TSV) extending vertically at least partially through the first die, a first dielectric layer disposed on the first die with a first metal bonding pad at least partially embedded in the first dielectric layer connected with the first TSV;

a second die having a second lateral footprint different from the first footprint, a second dielectric layer disposed on the second die with a second metal bonding pad at least partially embedded in the second dielectric layer, the first dielectric layer directly bonded to the second dielectric layer and the first metal bonding pad directly bonded to the second metal bonding pad; and an encapsulant disposed over a side surface of the first die, the encapsulant comprising glass, wherein the first metal bonding pad and the second metal bonding pad provide vertical electrical transmission between the first and second dies, and wherein at least a portion of the second metal bonding pad is disposed directly over the first TSV.

26. The microelectronic device of claim 25, wherein the first metal bonding pad connected with the first TSV comprises one or more of an electrical connection and a physical connection between the first metal bonding pad and the first TSV.

27. The microelectronic device of claim 25, wherein the encapsulant is present in a region laterally adjacent to the side surface of the first die.

28. The microelectronic device of claim 25, wherein the encapsulant directly contacts the side surface of the first die.

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